List of life sciences

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"Life Sciences" and "Bioscience" redirect here. For the scientific journals, see Life Sciences (journal) and BioScience.

Different kinds of living creatures studied in life sciences

- top: microorganisms (*E. coli* bacteria) and an animal (Goliath beetle)
- bottom: a plant (tree fern) and humans

This **list of life sciences** comprises the branches of science that involve the scientific study of life — such as animals (including human beings), microorganisms, and plants. This is one of the two major branches of natural science, the other being physical science, which is concerned with non-living matter. Biology is the overall natural science that studies life, with the other life sciences as its sub-disciplines.

Some life sciences focus on a specific type of organism. For example, zoology is the study of animals, while botany is the study of plants. Other life sciences focus on aspects common to all or many life forms, such as anatomy and genetics. Some focus on the micro scale (e.g., molecular biology, biochemistry), while others focus on larger scales (e.g., cytology, immunology, ethology, pharmacy, ecology). Another major branch of life sciences involves understanding the mind—neuroscience. Life-science discoveries are helpful in improving the quality and standard of life and have applications in health, agriculture, medicine, and the pharmaceutical and food science industries. For example, they have provided information on certain diseases, which has helped in the understanding of human health.[1]

Basic life science branches [edit]

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Main article: Outline of biology § Branches

- Biology – scientific study of life[2][3][4]
- Anatomy – study of form and function, in plants, animals, and other organisms[5]
  - Histology – the study of tissues
  - Neuroscience – the study of the nervous system
- Astrobiology – the study of the formation and presence of life in the universe[6]
- Biotechnology – study of combination of both the living organism and technology[7]
- Biochemistry – the study of the chemical reactions required for life to exist and function, usually focused on the cellular level[8]
  - Quantum biology – the study of quantum phenomena in organisms
- Bioinformatics – developing of methods or software tools for storing, retrieving, organizing and analyzing biological data to generate useful biological knowledge[9]
- Biophysics – study of biological processes by applying the theories and methods that have been traditionally used in the physical sciences[10]
  - Biomechanics – the study of the mechanics of living beings[11]
- Botany – study of plants[12]
  - Agrostology – the study of grasses and grass-like species
  - Dendrology - the study of woody plants and trees
  - Phycology – the study of algae[13]
- Cell biology (cytology) – study of the cell as a complete unit, and the molecular and chemical interactions that occur within a living cell[14]
- Developmental biology – the study of the processes through which an organism forms, from zygote to full structure
- Ecology – study of the interactions of living organisms with one another and with the non-living elements of their environment[15]
- Enzymology – study of enzymes
- Evolutionary biology – study of the origin and descent of species over time[16]
  - Evolutionary developmental biology – the study of the evolution of development including its molecular control
- Genetics – the study of genes and heredity
- Immunology – the study of the immune system[17]
- Marine biology – the study of ocean organisms
  - Biological oceanography – the study of life in the oceans and their interaction with the environment
- Microbiology – the study of microscopic organisms (microorganisms) and their interactions with other living organisms
  - Aerobiology – study of the movement and transportation of microorganisms in the air[18]
  - Bacteriology – study of bacteria
  - Virology – study of viruses and virus-like agents
- Molecular biology – the study of biology and biological functions at the molecular level, some cross over with biochemistry, genetics, and microbiology
  - Structural biology – a branch of molecular biology, biochemistry, and biophysics concerned with the molecular structure of biological macro-molecules
- Mycology – the study of fungi
- Paleobiology – the study of prehistoric organisms
- Parasitology – the study of parasites, their hosts, and the relationship between them
- Pathology – study of the causes and effects of disease or injury
- Human biology – the biological study of human beings
  - Pharmacology – study of drug action
  - Biological (or physical) anthropology – the study of humans, non-human primates, and hominids
  - Biolinguistics – the study of the biology and evolution of language
- Physiology – the study of the functioning of living organisms and the organs and parts of living organisms
- Population biology – the study of groups of conspecific organisms
  - Population dynamics – the study of short-term and long-term changes in the size and age composition of populations, and the biological and environmental processes influencing those changes. Population dynamics deals with the way populations are affected by birth and death rates, and by immigration and emigration, and studies topics such as ageing populations or population decline.
- Synthetic biology – the design and construction of new biological entities such as enzymes, genetic circuits and cells, or the redesign of existing biological systems
- Systems biology – the study of the integration and dependencies of various components within a biological system, with particular focus upon the role of metabolic pathways and cell-signaling strategies in physiology
- Theoretical biology – use of abstractions and mathematical models to study biological phenomena
- Toxicology – the study of poisons
- Zoology – the study of (generally non-human) animals
  - Ethology – the study of animal behavior[19]

Applied life science branches and derived concepts [edit]

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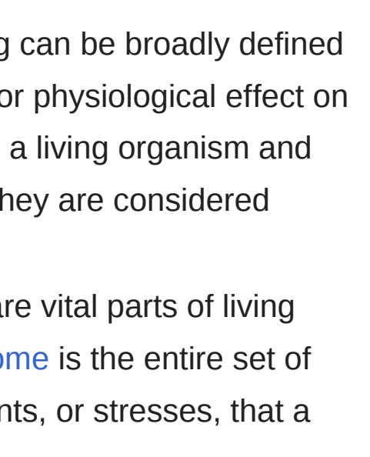
- Agriculture – science and practice of cultivating plants and livestock
  - Agronomy – science of cultivating plants for resources
- Biocomputers – systems of biologically derived molecules, such as DNA and proteins, are used to perform computational calculations involving storing, retrieving, and processing data. The development of biological computing has been made possible by the expanding new science of nanobiotechnology.[20]
- Biocontrol – bioeffector-method of controlling pests (including insects, mites, weeds and plant diseases) using other living organisms.[21]
- Bioengineering – study of biology through the means of engineering with an emphasis on applied knowledge and especially related to biotechnology
- Bioelectronics – field at the convergence of electronics and biological sciences. The electrical state of biological matter significantly affects its structure and function, compare for instance the membrane potential, the signal transduction by neurons, the isoelectric point (IEP) and so on. Micro- and nano-electronic components and devices have increasingly been combined with biological systems like medical implants, biosensors, lab-on-a-chip devices etc. causing the emergence of this new scientific field.[22]
- Biomaterials – any matter, surface, or construct that interacts with biological systems. As a science, biomaterials is about fifty years old. The study of biomaterials is called biomaterials science. It has experienced steady and strong growth over its history, with many companies investing large amounts of money into the development of new products. Biomaterials science encompasses elements of medicine, biology, chemistry, tissue engineering and materials science.
- Biomedical science – healthcare science, also known as biomedical science, is a set of applied sciences applying portions of natural science or formal science, or both, to develop knowledge, interventions, or technology of use in healthcare or public health. Such disciplines as medical microbiology, clinical virology, clinical epidemiology, genetic epidemiology and pathophysiology are medical sciences.
- Biomonitoring – measurement of the body burden of toxic chemical compounds, elements, or their metabolites, in biological substances.[23][24] Often, these measurements are done in blood and urine.[25]
- Biopolymer – polymers produced by living organisms; in other words, they are polymeric biomolecules. Since they are polymers, biopolymers contain monomeric units that are covalently bonded to form larger structures. There are three main classes of biopolymers, classified according to the monomeric units used and the structure of the biopolymer formed: polynucleotides (RNA and DNA), which are long polymers composed of 13 or more nucleotide monomers; polypeptides, which are short polymers of amino acids; and polysaccharides, which are often linear bonded polymeric carbohydrate structures.[26][27][28]
- Biotechnology – manipulation of living matter, including genetic modification and synthetic biology[29]
- Conservation biology – the management of nature and of Earth's biodiversity with the aim of protecting species, their habitats, and ecosystems from excessive rates of extinction and the erosion of biotic interactions. It is an interdisciplinary subject drawing on natural and social sciences, and the practice of natural resource management.[30]
- Environmental health – multidisciplinary field concerned with environmental epidemiology, toxicology, and exposure science.
- Fermentation technology – study of use of microorganisms for industrial manufacturing of various products like vitamins, amino acids, antibiotics, beer, wine, etc.[31]
- Food science – applied science devoted to the study of food. Activities of food scientists include the development of new food products, design of processes to produce and conserve these foods, choice of packaging materials, shelf-life studies, study of the effects of food on the human body, sensory evaluation of products using panels or potential consumers, as well as microbiological, physical (texture and rheology) and chemical testing.[32][33][34]
- Genomics – application of recombinant DNA, DNA sequencing methods, and bioinformatics to sequence, assemble, and analyze the function and structure of genomes (the *complete* set of DNA within a single cell of an organism).[35][36] The field includes efforts to determine the entire DNA sequence of organisms and fine-scale genetic mapping. The field also includes studies of intragenomic phenomena such as heterosis, epistasis, pleiotropy and other interactions between loci and alleles within the genome.[37] In contrast, the investigation of the roles and functions of single genes is a primary focus of molecular biology or genetics and is a common topic of modern medical and biological research. Research of single genes does not fall into the definition of genomics unless the aim of this genetic, pathway, and functional information analysis is to elucidate its effect on, place in, and response to the entire genome's networks.[38][39]
- Health sciences – sciences which focus on health, or health care, as core parts of their subject matter. These two subject matters relate to multiple academic disciplines, both STEM disciplines, as well as emerging patient safety disciplines (such as social care research), and are both relevant to current health science knowledge.
  - Medical devices – A medical device is an instrument, apparatus, implant, in vitro reagent, or similar or related article that is used to diagnose, prevent, or treat disease or other conditions, and does not achieve its purposes through chemical action within or on the body (which would make it a drug).[40] Whereas *medicinal products* (also called *pharmaceuticals*) achieve their principal action by pharmacological, metabolic or immunological means, *medical devices* act by other means like physical, mechanical, or thermal means.
  - Medical imaging – the technique and process used to create images of the human body (or parts and function thereof) for clinical or physiological research purposes[41]
  - Immunotherapy – the "treatment of disease by inducing, enhancing, or suppressing an immune response".[42] Immunotherapies designed to elicit or amplify an immune response are classified as activation immunotherapies, while immunotherapies that reduce or suppress are classified as suppression immunotherapies.[43]
- Kinesiology – scientific study of human movement. Kinesiology, also known as human kinetics, addresses physiological, mechanical, and psychological mechanisms. Applications of kinesiology to human health include: biomechanics and orthopedics; strength and conditioning; sport psychology; methods of rehabilitation, such as physical and occupational therapy; and sport and exercise. Individuals who have earned degrees in kinesiology can work in research, the fitness industry, clinical settings, and in industrial environments.[44] Studies of human and animal motion include measures from motion tracking systems, electrophysiology of muscle and brain activity, various methods for monitoring physiological function, and other behavioral and cognitive research techniques.[45]
- Optogenetics – a neuromodulation technique employed in neuroscience that uses a combination of techniques from optics and genetics to control and monitor the activities of individual neurons in living tissue—even within freely-moving animals—and to precisely measure the effects of those manipulations in real-time.[46] The key reagents used in optogenetics are light-sensitive proteins. Spatially-precise neuronal control is achieved using optogenetic actuators like channelrhodopsin, halorhodopsin, and archaerhodopsin, while temporally-precise recordings can be made with the help of optogenetic sensors like Clomeleon, Mermaid, and SuperClomeleon.[47]



Parasagittal MRI of the head, with aliasing artifacts

- Pharmacogenomics – field of science and technology that analyses how genetic makeup affects an individual's response to drugs.[48] Pharmacogenomics (a portmanteau of pharmacology and genomics) deals with the influence of genetic variation on drug response in patients by correlating gene expression or single-nucleotide polymorphisms with a drug's efficacy or toxicity.[49]
- Pharmacology – branch of medicine and biology concerned with the study of drug action,[50] where a drug can be broadly defined as any human-made, natural, or endogenous (within the body) molecule which exerts a biochemical and/or physiological effect on the cell, tissue, organ, or organism. More specifically, it is the study of the interactions that occur between a living organism and chemicals that affect normal or abnormal biochemical function. If substances have medicinal properties, they are considered pharmaceuticals.
- Proteomics – the large-scale study of proteins, particularly their structures and functions.[51][52] Proteins are vital parts of living organisms, as they are the main components of the physiological metabolic pathways of cells. The proteome is the entire set of proteins,[53] produced or modified by an organism or system. This varies with time and distinct requirements, or stresses, that a cell or organism undergoes.

See also [edit]

- Outline of biology
- Divisions of pharmacology
- Control theory

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Further reading [edit]

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